FLEET TEAM DIGEST

737MAX-FTD-23-18003

Issue Title : Intermittent Iridium SATCOM Datalink Outages

Airplane Model	ATA	Minor Model(s)
737MAX	2315-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
08/14/2018	04/17/2020	04/13/2020
Estimated Completion Date	**Next Update Date**	

Revision Description

Status and Final Action sections have been updated to reflect publication of the Service Bulletin and Service Letter.

Applicability

737MAX airplanes with the ICS-300 SATCOM system installed.

Description

Operators reported intermittent Iridium ICS-300 SATCOM data link outages when flying out of VHF range, specifically over oceanic regions. The outages result in a datalink 'NO COMM' condition. The 'NO COMM' message displays on the MCDU. Outages have typically been reported to be between 10 seconds to 15 minutes. Two outages have been as long as 1 hour.

Boeing, Collins Aerospace and Iridium are investigating the flight and SDU BITE data to determine the root cause. This condition is currently believed to be unique to the Rockwell ICS-300 system on the 737 airplane.

Background

The issue has been reported to occur only when the airplane has flown out of VHF range and SATCOM becomes the only datalink communication medium available. Certain flight routes such as Future Air Navigation System (FANS) routes require the SATCOM datalink capability. VHF datalink works normally while the airplane is within VHF range.

Since the airplane is out of VHF range and the SATCOM system is momentarily inop, the Communications Management Unit (CMU) will enter a 'NO COMM' state and display the message on the MCDU.

There is no Maintenance Messages or Status Messages associated with these intermittent outages.

Status

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Boeing began a Service Related Problem (SRP) project 737MAX-SRP-23-0092 to track this issue for resolution.

Boeing, Collins and Iridium are conducting a root cause analysis on the noted condition. Data has been continually collected from operators who have been experiencing the outages. Several potential contributing factors have been analyzed and tested for.

Boeing extensively tested the ICS-300 system in the labs for continuous operation and did not see similar outages reported.

Collins also continually tested the ICS-300 in several configurations and did not see similar outages. The testing included a Satellite Data Unit (SDU) that had previously experienced outages in service.

Outages were reproduced in the Collins Labs when significant signal attenuation was introduced. The level of attenuation introduced however was several dB higher than the theoretical level expected to be experienced on the airplane.

Boeing has performed additional testing with Collins to collect data on planes that have previously experienced outages. Multiple root causes and potential contributing factors have been ruled out.

Boeing, Collins and Iridium have identified an improvement with the Iridium modem and are working to have it released to the fleet. Iridium has completed their new modem firmware as of March 2019. Collins Aerospace is now going through regression testing to certify the new modem upgrade. Current testing results show considerable improvement.

As of September 2019, Collins Aerospace and Boeing testing of the updated firmware resulted in improvements in the amount of time the satellite signal strength was at max levels, as well as a higher percentage of uplink acknowledgements.

When compared to the previous versions of the modem firmware that are not experiencing this issue, Iridium identified parameters in the satellite acquisition algorithms that were not optimized for the ICS-300 system. Also, the battery save mode was linked to the root cause, and hence, it is not enabled/used in the updated firmware.

Iridium released a firmware update with more aggressive satellite acquisition protocol to incorporate into the SDU. Collins has completed regression testing of the new SDU configuration, and final documentation is being reviewed by Collins, Collins' TCR, and Boeing SME's/AR's before submitting package for FAA approval (minor change). New configuration will be tracked at the mod level – no part number change.

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Collins Aerospace has published Service Bulletin ICS-300-23-1 - 'Communications - ICS-300 -Firmware Upgrade' (publication 523-0825994) on 13 March 2020.

Boeing has published Service Letter 737-SL-23-061A - 'Intermittent Iridium ICS-300 SATCOM Datalink' Outages on 13 April 2020.

Interim Action

No interim or mitigating actions have been determined at this time.

In the meantime, operators flying routes out of VHF range with the ICS-300 SATCOM system are encouraged to report their experience.

Final Action

The list of impacted unit serial numbers is included in Section 1.A. of the Service Bulletin.

Operators will need to ship SDU's to Collins' Service Centers to incorporate Service Bulletin 523-0825994 for SDU firmware upgrade. This service bulletin may be installed only by a Rockwell Collins representative or an authorized Rockwell Collins Service Center.

Milestones

13 March 2020: Publication of Service Bulletin ICS-300-23-1 - 'Communications - ICS-300 -Firmware Upgrade' (publication 523-0825994)

13 April 2020: Publication of Service Letter 737-SL-23-061A - 'Intermittent Iridium ICS-300 SATCOM Datalink

Reference Categories

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Reference Type	Reference Number	Revision Number	Reference Date
Service Related Problem (SRP)	737MAX-SRP-23-0092		
Service Letter (SL)	737-SL-23-061	Initial	

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Satellite Data Unit (SDU)			701-10300-00

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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